Ensemble-Based SIDM Weak-Lensing Forecasts for Spec-S5 White-Paper Use

March 20, 2026

1 Executive Summary

1.1 Dwarf lenses

Building on Yi-Fei Luo's single-halo SIDM forecasting baseline, we implement an ensemble-level forecast in which dwarf lenses are sampled as a stellar-mass-selected population and propagated through SHMR scatter, halo concentration scatter, profile projection, and stacked $\Delta\Sigma(R)$. The key result is methodological: the SIDM–CDM contrast is evaluated for a *distribution* of halo properties rather than one fiducial halo, which is a better match to the stacked-lensing observable expected from Spec-S5 + Stage-IV imaging.

1.2 Cluster lenses

For clusters, we implement a halo-mass-selected ensemble forecast and include Tier-2 hybrid profile modeling (SIDM-modified inner halo plus DK14-like outer structure). This provides a realistic baseline for stacked $\Delta\Sigma$ comparisons, but the scientific interpretation remains systematics-limited: off-centering, concentration-population effects, and especially baryonic physics can generate profile changes that are degenerate with SIDM-induced modifications over overlapping radial ranges.

2 Motivation and Scope

The science goal is to test whether the expanded spectroscopic lens samples from Spec-S5, combined with Stage-IV weak lensing, can distinguish CDM from SIDM using stacked galaxy–galaxy lensing signals in dwarf and cluster regimes. The present implementation is intentionally a controlled first-order forecast:

- no 2-halo term,
- no explicit off-centering model,
- no baryonic response model,
- no full survey covariance treatment.

These omissions are deliberate for transparency and speed; they define the caveats on absolute interpretability.

3 Method Summary (Main Text)

3.1 Halo ensemble extension

The primary contribution of this work is replacing a single-fiducial-halo calculation with a sampled halo ensemble:

- **HMF mode (clusters):** sample from dn/dM with selection and redshift distributions.
- **SHMR mode (dwarfs):** sample stellar masses, map to halo masses via SHMR with scatter, and apply concentration scatter.

Profiles are generated halo-by-halo, projected to $\Sigma(R)$ and $\Delta\Sigma(R)$, interpolated to common radial bins, and stacked.

3.2 Profile model assumptions

SIDM inner profiles are generated with the `parametricSIDM` wrapper. Tier-2 uses a DK14-like outer profile attachment and smooth stitching to avoid projection artifacts. In the checked-in default configs used for the summary figures, Tier-3 empirical outskirts correction is disabled.

3.3 Projected lensing and toy distinguishability

The local projection implementation computes

$$\Sigma(R) = 2\int_0^{z_{\max}} \rho\left(\sqrt{R^2+z^2}\right)\, dz, \tag{1}$$

$$\bar{\Sigma}(<R) = \frac{2}{R^2}\int_0^R \Sigma(R')R'\, dR', \tag{2}$$

$$\Delta\Sigma(R) = \bar{\Sigma}(<R) - \Sigma(R), \tag{3}$$

with log-space interpolation and stable numerical quadrature. Distinguishability is summarized with a toy

$$\Delta\chi^2 = \sum_i \frac{\left[\Delta\Sigma_i^{\mathrm{SIDM}} - \Delta\Sigma_i^{\mathrm{CDM}}\right]^2}{\sigma_i^2}, \tag{4}$$

using simple fractional-error scenarios.

4 Representative Forecast Products

5 Findings and Caveats for White-Paper Use

Findings.

- The ensemble treatment is technically stable and reproducible (median-mass, radial-bin, fixed-seed, and single-halo-limit checks pass in the default cluster and dwarf runs).
- The stacked SIDM/CDM contrast is preserved after introducing realistic halo-population breadth, so the forecast is not an artifact of a single chosen halo.

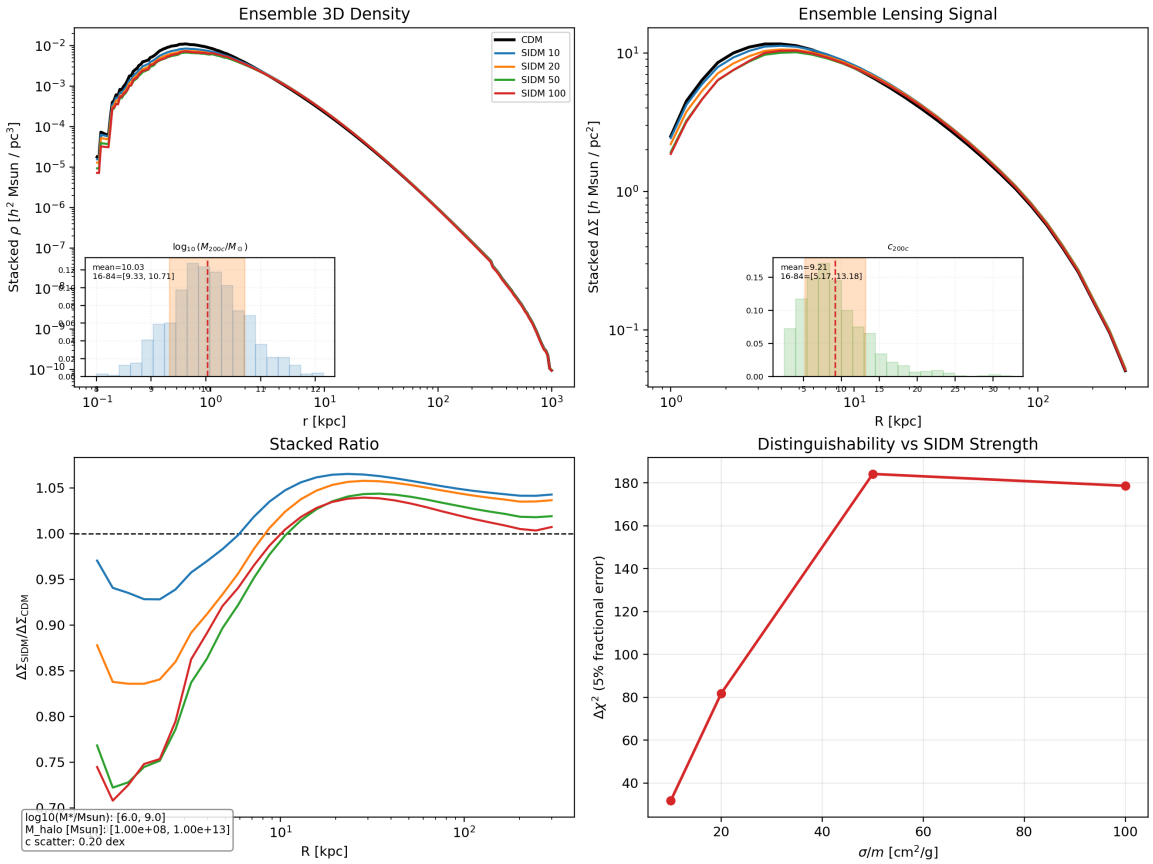

Figure 1: **Dwarf ensemble summary (SHMR-selected sample).** Multi-panel diagnostic from the current default dwarf configuration (`docs/dwarf_ensemble_config.yaml`). The density panel and lensing panel show stacked CDM and SIDM predictions over the benchmark high-σ/m grid; the ratio panel shows SIDM/CDM in projected space; the $\Delta\chi^2$ panel gives toy distinguishability under the assumed fractional-error model. Inset histograms report the sampled halo-mass and concentration distributions (including mean and 16–84% ranges), making explicit that the forecast is population-based rather than single-halo. Units follow the project convention: stacked ρ in $h^2\, M_\odot\, \mathrm{pc}^{-3}$ and stacked $\Delta\Sigma$ in $h\, M_\odot\, \mathrm{pc}^{-2}$.

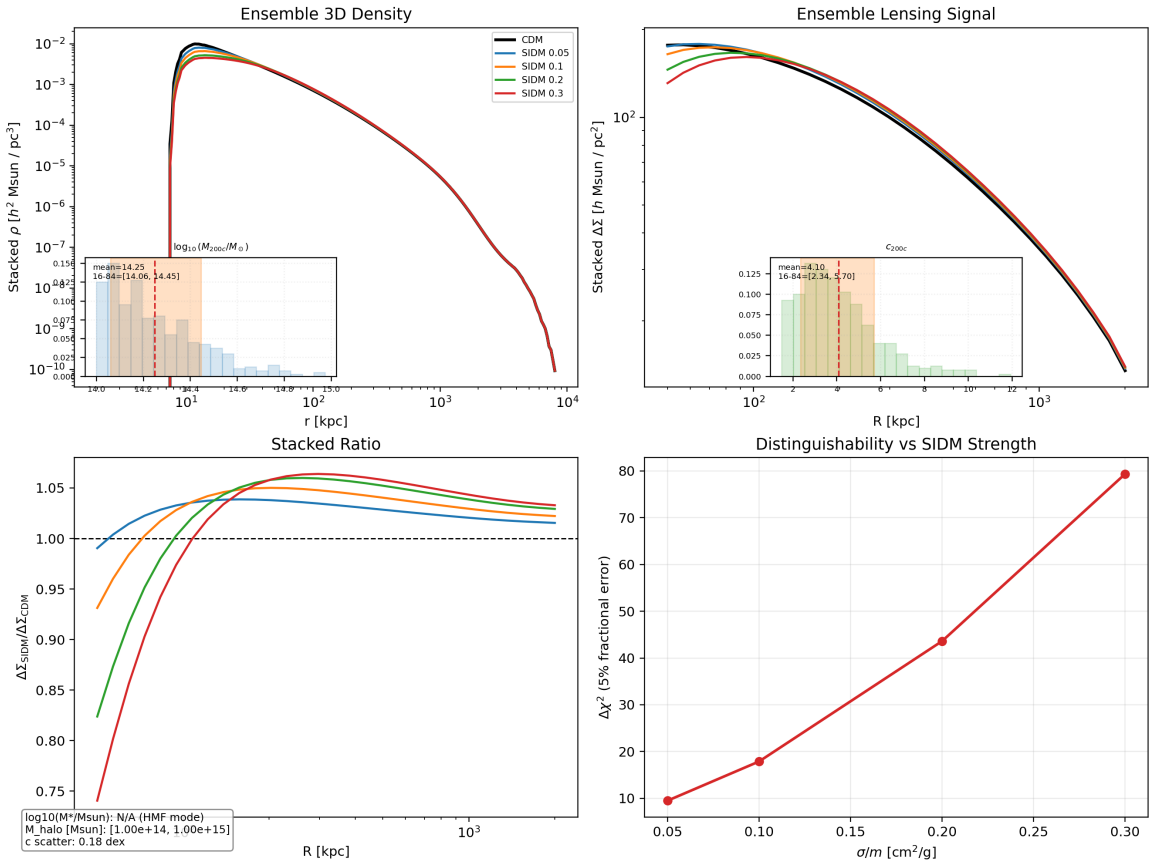

Figure 2: **Cluster ensemble summary (HMF-selected sample).** Multi-panel diagnostic from the current default cluster configuration (`docs/cluster_ensemble_config.yaml`), with Tier-2 hybrid profile treatment enabled. The figure summarizes stacked profile behavior over the benchmark sub-unity SIDM grid and reports toy $\Delta\chi^2$ sensitivity. As in the dwarf case, inset panels document the realized halo-mass and concentration distributions used in stacking. Interpretation caveat: cluster-side SIDM-like variations in $\Delta\Sigma$ can be degenerate with off-centering, concentration-selection effects, and baryonic profile modifications; these systematics are not yet jointly modeled in this first-order pipeline.

- Tier-2 outer-profile attachment changes the large-radius baseline while retaining inner SIDM response, enabling more realistic cluster-scale profile comparisons.

Caveats.

- The reported $\Delta\chi^2$ values are toy metrics tied to assumed fractional errors, not final survey-significance claims.
- Cluster interpretation is likely dominated by degeneracies with off-centering, concentration-population effects, and baryonic physics.
- No 2-halo term, full covariance, or full baryonic nuisance framework is included at this stage.

A Appendix: Reproducibility Details

A.1 Configuration files

- Cluster baseline config: `docs/cluster_ensemble_config.yaml`
- Dwarf baseline config: `docs/dwarf_ensemble_config.yaml`

Key default settings:

- Cluster: $N_{\rm halos} = 400$, $M_{200c} \in [10^{14}, 10^{15}]\, M_\odot$, $z \sim \mathcal{N}(0.45, 0.15)$ truncated to $[0.2, 0.8]$, $\sigma_{\log c} = 0.18$, $\sigma/m = \{0.05, 0.1, 0.2, 0.3\}$ cm^2 g^{-1}.
- Dwarf: $N_{\rm halos} = 800$, $\log_{10} M_\star \sim \mathcal{N}(7.5, 0.5)$ truncated to $[6, 9]$, SHMR scatter 0.3 dex, $\sigma_{\log c} = 0.20$, $\sigma/m = \{10, 20, 50, 100\}$ cm^2 g^{-1}.

A.2 Run commands

```
python scripts/run_ensemble_forecast.py \
  --config-path docs/cluster_ensemble_config.yaml \
  --output-root outputs

python scripts/run_ensemble_forecast.py \
  --config-path docs/dwarf_ensemble_config.yaml \
  --output-root outputs
```

A.3 Primary outputs used in this note

- `outputs/figures/dwarf_ensemble_summary.png`
- `outputs/figures/cluster_ensemble_summary.png`
- `outputs/tables/dwarf_ensemble_delta_chi2_summary.csv`
- `outputs/tables/cluster_ensemble_delta_chi2_summary.csv`
- `outputs/tables/*_ensemble_validation_checks.csv`

A.4 Code modules

- Ensemble sampling: `src/sidm_stagev_forecast/ensemble.py`
- YAML normalization: `src/sidm_stagev_forecast/ensemble_yaml.py`
- Profile construction: `src/sidm_stagev_forecast/profiles.py`
- Outer profile/stitching: `src/sidm_stagev_forecast/outer_profiles.py`, `stitch.py`
- Projection: `src/sidm_stagev_forecast/projection.py`
- Stacking/forecast metrics: `src/sidm_stagev_forecast/stacking.py`, `forecast.py`
- Pipeline driver: `scripts/run_ensemble_forecast.py`